Vaishnavi Shivkumar

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EDUCATION

International Institute of Information Technology, Hyderabad 2022 – 2026
B.Tech in Electronics and Communication Engineering (Honours) Hyderabad, India

- **CGPA:** 8.51/10; **Research CGPA:** 9.85/10; **Dean's Merit List:** Spring 2024, Monsoon 2024, Spring 2025.
- **Teaching/Service:** TA for Responsible and Safe AI Systems, Intro to Statistical Signal Processing, Introduction to Film Studies; Chief Editor, Student Media Body; Apex Council Head, Student Mentorship Body; PR Lead, Student Parliament.

PUBLICATIONS & BLOGS

- P. Kodali, *V. Shivkumar*, S. Joshi, M. Choudhary, P. Kumaraguru, M. Shrivastava. "Adapting Multilingual Models to Code-Mixed Tasks via Model Merging." **CODS 2025**. arXiv
- K. Aravindan, A. Rastogi, A. Prasad, K. Aneja, S. Jain, *V. Shivkumar*, P. Kumaraguru. "OPIUM: Mitigating Steering Externalities and Over-Refusal via Dual Objective Latent Optimization." **ICML 2026 Mech Interp Workshop**. OpenReview
- R. Bhattacharya, H. Shankar, *V. Shivkumar*, P. Kumaraguru. "MetaGMT: Improving Actionable Interpretability of Graph Multilinear Networks via Meta-Learning Filtration." **preprint**, 2025. arXiv

EXPERIENCE

Mastercard AI Garage 05/2026 – Present
Data Scientist - 1 Gurugram, India

- Worked on the ASI 2.0 product, from data pull, to model retraining, to packaging, and platform migration for deployment of the product worldwide

Mastercard AI Garage 05/2025 – 07/2025
Research Intern Gurugram, India

- Worked on graph-based financial fraud detection; achieved 106% improvement in fraud GDR; **US Patent filed**; received Pre-Placement Offer.

Celebal Technologies 06/2023 – 07/2023
Machine Learning Intern Remote

- Built ML-based NILM and early sensor fault/anomaly detection models using XGBoost, Random Forests, and SVMs.

RESEARCH

Precog Lab, IIIT Hyderabad 05/2024 – Present
Undergraduate Researcher; advised by Prof. Ponnurangam Kumaraguru Hyderabad, India

- **NLP:** Code-mixed model merging, representation alignment/drift, multilingual robustness, distribution shift, and catastrophic forgetting.
- **Federated Learning:** Client-level fairness and allocative harms under heterogeneous resource constraints; binarized factor models for efficient learning.
- **Video/World Models:** Designing probes for latent symbolic/rule reasoning from continuous trajectories; direction connected to reasoning/world-model work with Prof. Makarand Tapaswi.
- **Autonomous Driving:** Created a counterfactual grounding benchmark for Autonomous Driving pedestrian settings (***submitted to ICRA 2027***)
- **Collaborations:** Worked with researchers from LTRC-IIITH, MBZUAI, IIT Madras, and UIUC across NLP and federated learning projects.

Signal Processing Research Centre, IIIT Hyderabad 01/2025 – 04/2025
Independent Research Project; advised by Prof. Santosh Nannuru Hyderabad, India

- Studied network inference from diffusion/SI dynamics and proposed a latent low-rank formulation for partially observed dynamical networks.

PROJECTS

Critical Evaluation of TinyStories Advanced NLP

- Audited TinyStories-style evaluation pipelines for dataset duplication, vocabulary-control leakage, and scoring artifacts; trained small transformers/probes to test compositional story understanding.

Multi-Agent Reinforcement Learning for E-Commerce Fulfillment Optimization for ML

- Built a multi-agent RL system using Graph Autoencoders and DQNs; integrated VRP solvers and benchmarked against heuristic baselines.

SKILLS

Languages/Tools: Python, PyTorch, SQL, MATLAB, C/C++, Bash, Verilog, LaTeX, XGBoost, scikit-learn
Areas: NLP, Graph ML, Deep Learning, Computer Vision, Federated Learning, Interpretability, Multimodality, AI Safety, Signal Processing, Statistical Inference, Data Engineering